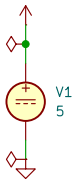
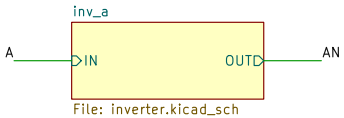
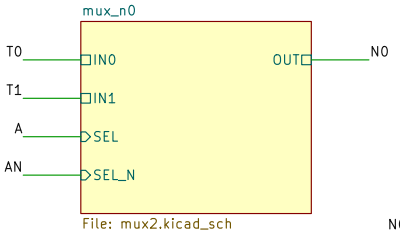
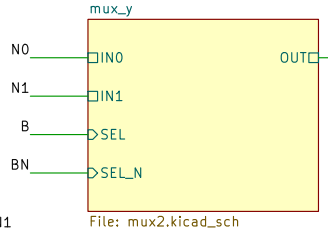
Look-up table with two inputs, built from transmission gates and SRAM cells
Y = T[2 B + A], table written through the bit lines

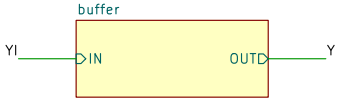
Supply



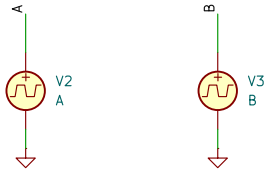


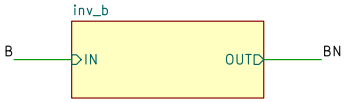


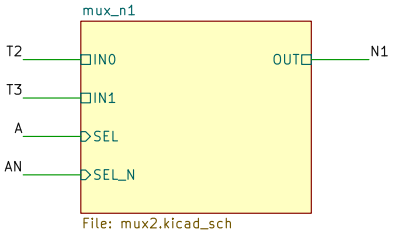




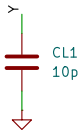
Inputs A and B



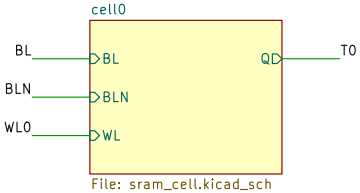


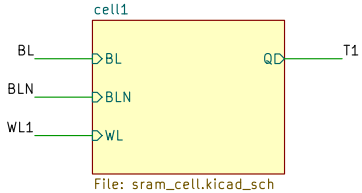


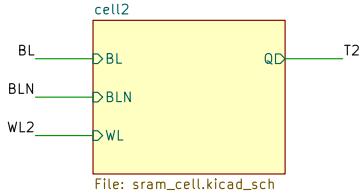
Load

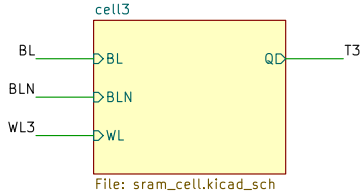


Configuration memory: four SRAM cells hold the table bits T0..T3

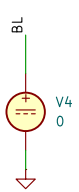


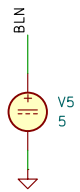


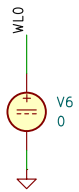


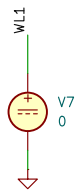


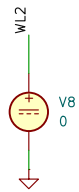
Bit lines BL, BLN (data) and word lines WL0..WL3 (address)

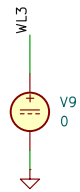




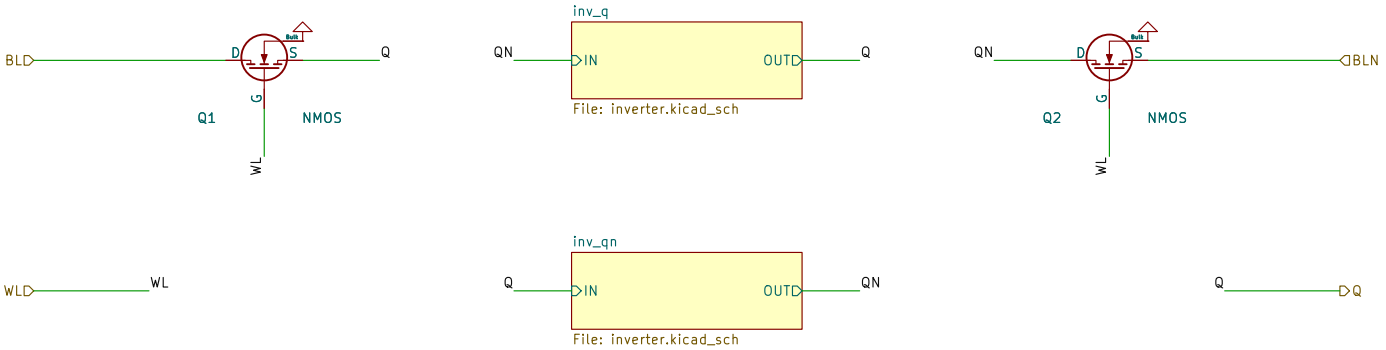




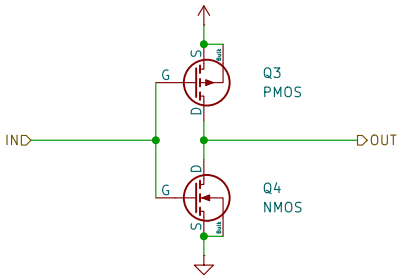




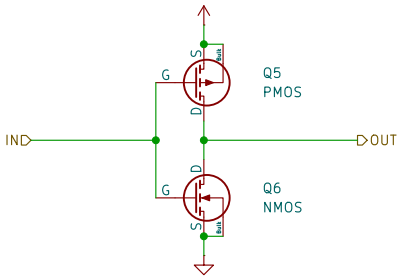
SRAM cell with six transistors.
Two inverters hold each other: Q = not QN, QN = not Q.
While WL = 1 the access transistors connect the cell to the bit lines,
and the bit lines, driven from outside, overwrite the stored value.



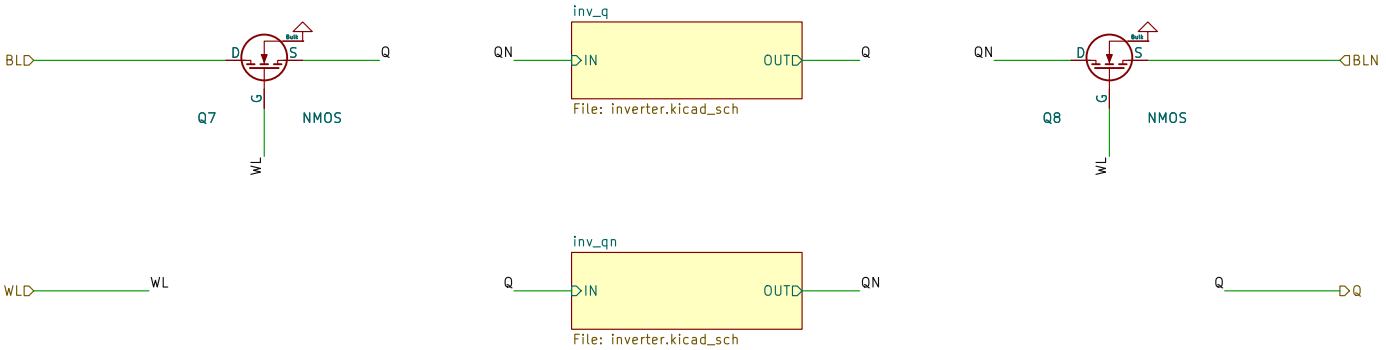
CMOS inverter: OUT = not IN
The PMOS conducts for IN = 0, the NMOS for IN = 1.



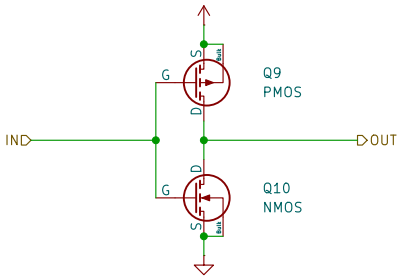
CMOS inverter: OUT = not IN
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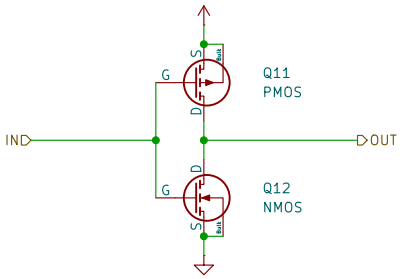
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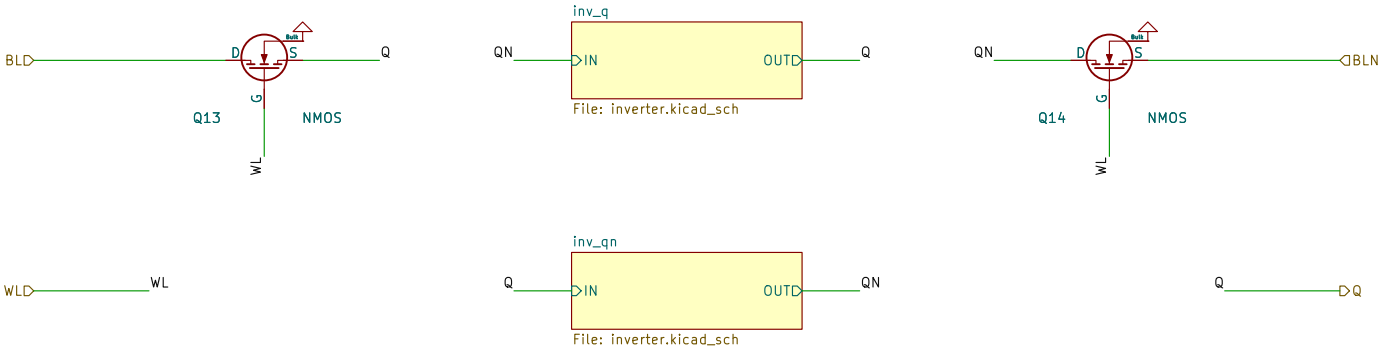
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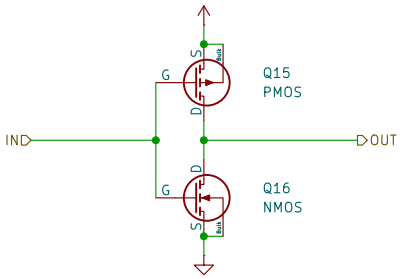
CMOS inverter: OUT = not IN
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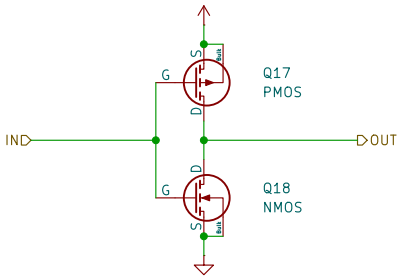
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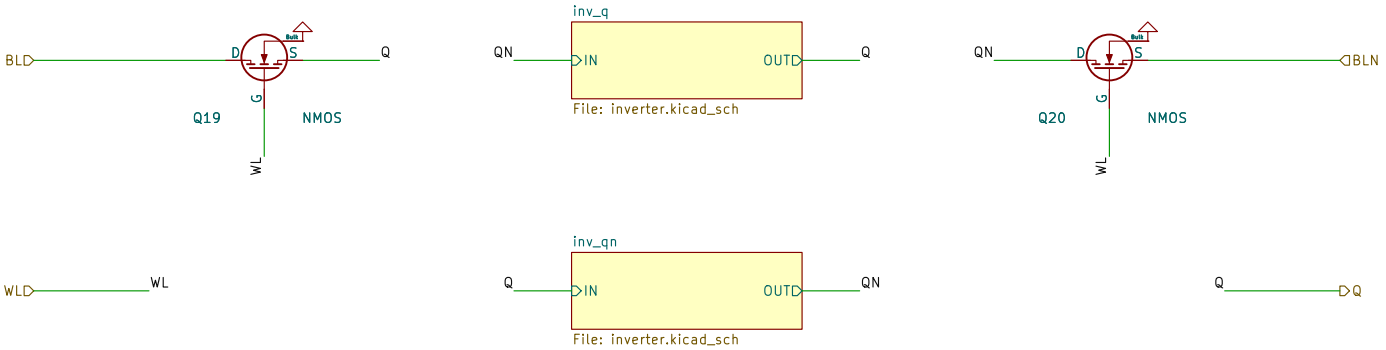
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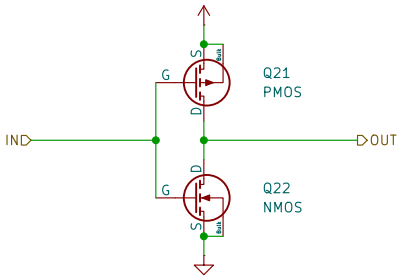
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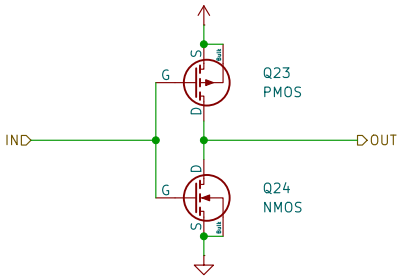
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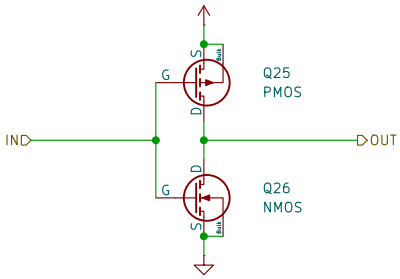
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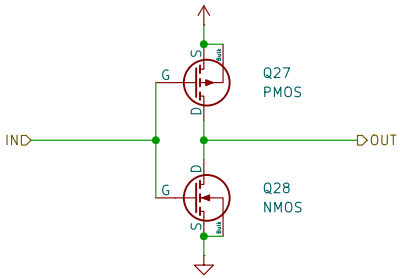
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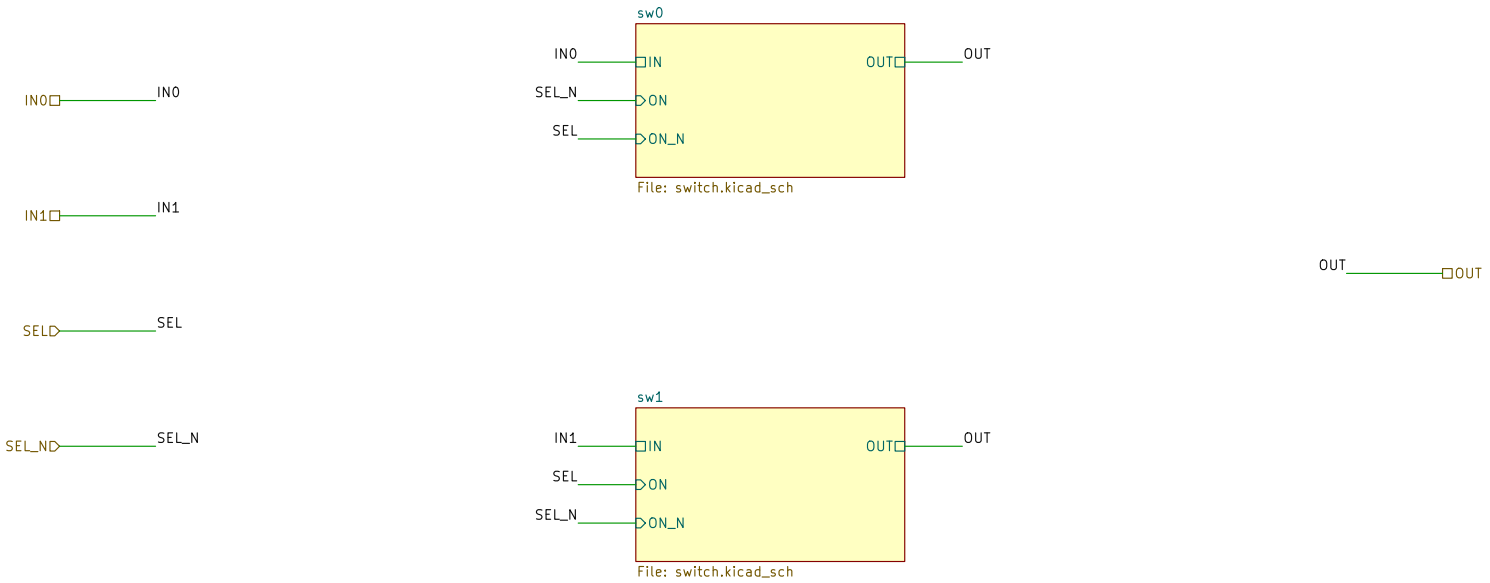
CMOS inverter: OUT = not IN
The PMOS conducts for IN = 0, the NMOS for IN = 1.



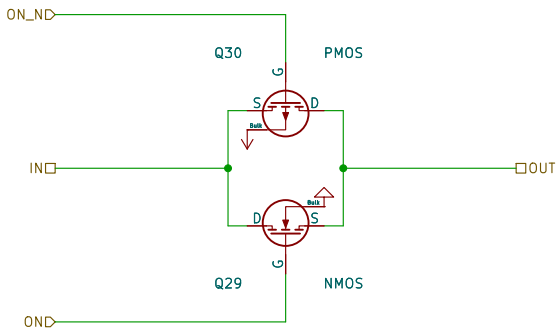
CMOS inverter: OUT = not IN
The PMOS conducts for IN = 0, the NMOS for IN = 1.



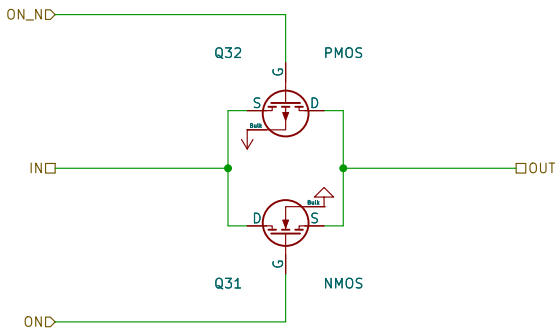
2:1 multiplexer: OUT = IN1 if SEL = 1, otherwise IN0.
Exactly one of the two switches is closed.



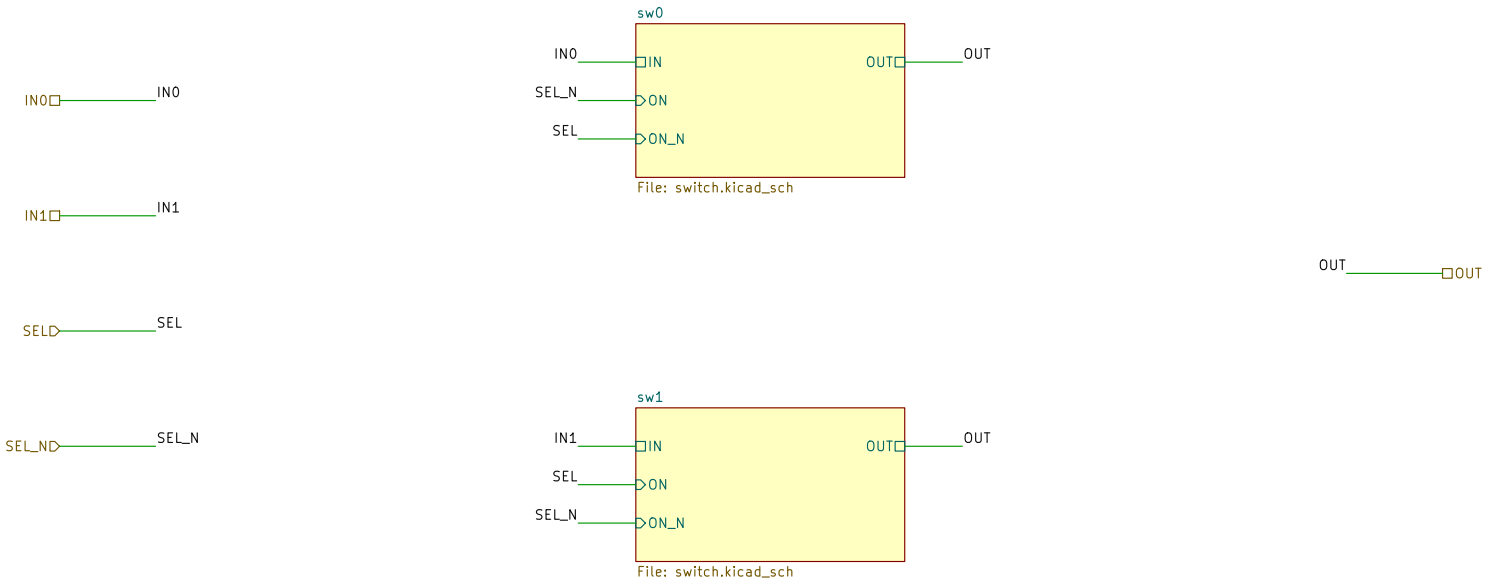
Transmission gate: IN and OUT are connected when ON = 1 and ON_N = 0.
The NMOS passes a 0 well, the PMOS passes a 1 well; together they pass both.



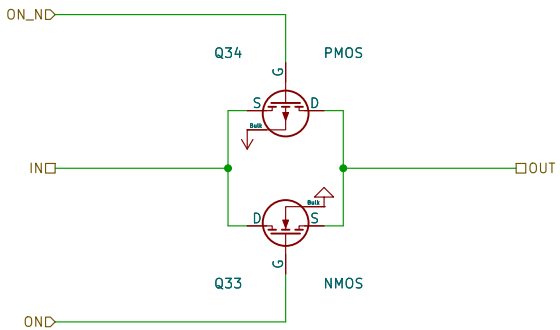
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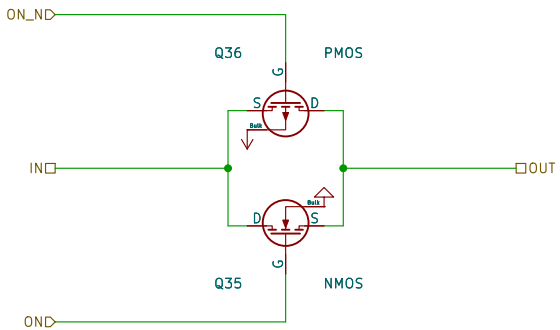
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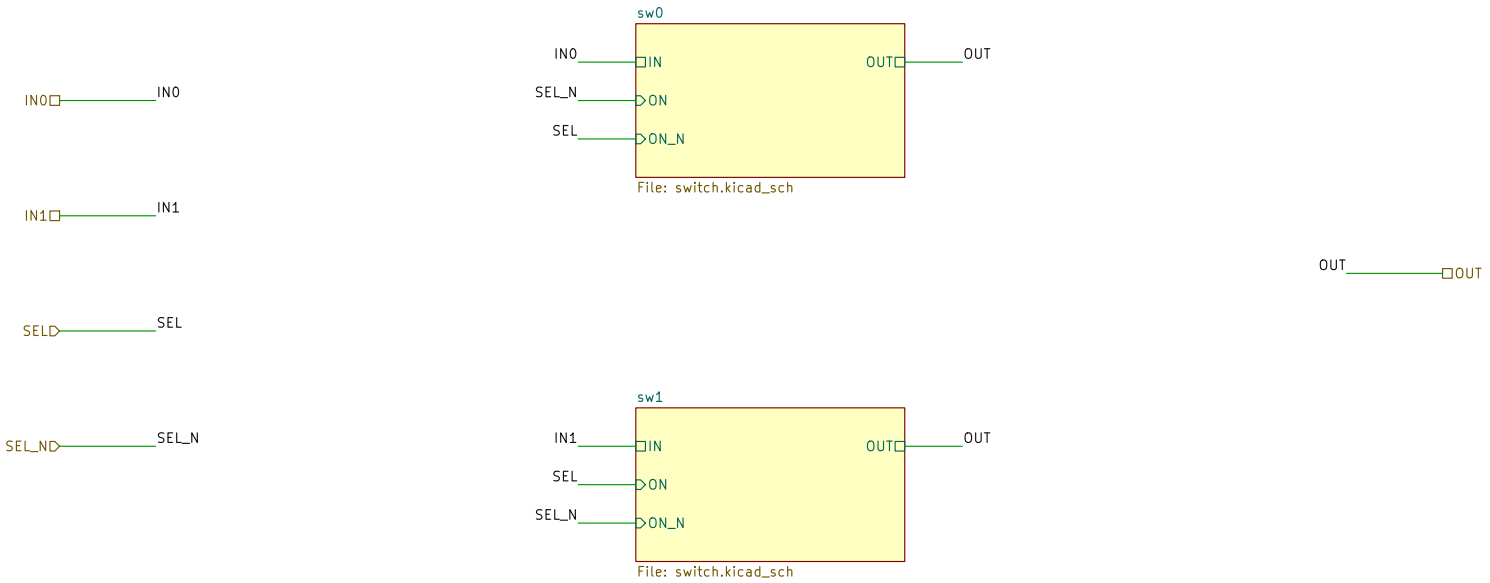
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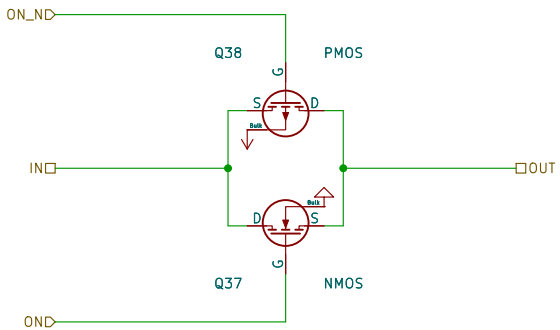
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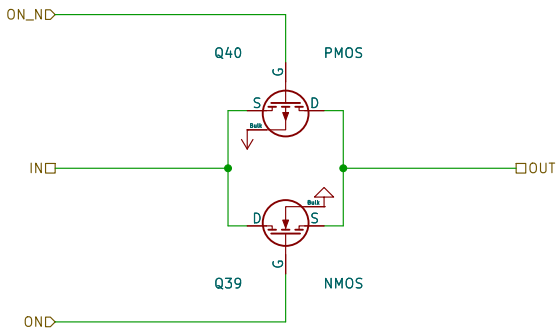
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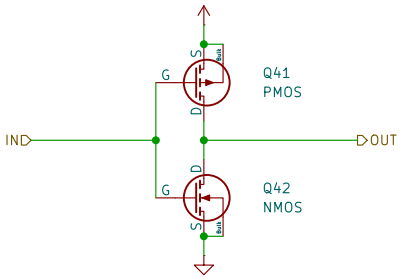
Transmission gate: IN and OUT are connected when ON = 1 and ON_N = 0.
The NMOS passes a 0 well, the PMOS passes a 1 well; together they pass both.



Output buffer: two inverters in series.
The first one restores the logic level, the second one restores the polarity.



CMOS inverter: OUT = not IN
The PMOS conducts for IN = 0, the NMOS for IN = 1.



CMOS inverter: OUT = not IN
The PMOS conducts for IN = 0, the NMOS for IN = 1.

